Differentiable simulation for agrivoltaic tracker control under a yield floor: what learning adds and what shade-response uncertainty costs

Quang-Vinh Dang*

British University Vietnam, Hung Yen, Vietnam

Abstract

Agrivoltaic single-axis trackers can rotate to let light through, but how much a learned controller adds over simple rules under a yield-retention floor, and how shade-response uncertainty limits design, are unknown. We built a differentiable simulator of tracker geometry, photovoltaic output, light and microclimate under the modules, and crop growth, and trained a neural controller through whole seasons by back-propagation. The controller and written rules were compared on 400 crop–site pairs, with each site's density chosen from its history and evaluation on held-out sites. Light-sharing rules allowed arrays 2.1–2.9 times as dense as static arrays and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24, but the floor was missed in 24–31% of seasons. The neural controller exceeded the written phenology rule by 0.044 LER and an optimised window by only 0.007, with no detectable difference at matched retention. Shade curves learned from 428 field observations were no better than proportional loss for staple crops, and the simulator was milder than the data for maize. A density chosen under the mean response left 70% of pairs at the floor when another plausible response held (oracle: 87%). Conformal margins attained the stated compliance within an ensemble of responses, a learned design network did worse than a simple margin, and only large margins were safe outside the ensemble. Differentiable simulation made control and design tractable, but data, not the learner, limit what can be guaranteed. The evaluation is by simulation on public data.

Keywords: Agrivoltaics, Differentiable simulation, Solar tracking, Neural control, Crop modelling, Conformal prediction, Decision under uncertainty

*Corresponding author.

Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

1. Introduction

Agrivoltaic (AV) systems put crops and photovoltaic (PV) modules on the same land, so that a hectare can deliver food and electricity together (Dupraz et al., 2011; Dinesh and Pearce, 2016). Land-equivalent ratios (LER) above one have been reported in field trials and models (Amaducci et al., 2018; Valle et al., 2017; Trommsdorff et al., 2021), and partial shade can lower canopy temperature and conserve soil water (Marrou et al., 2013; Hassanpour Adeh et al., 2018; Weselek et al., 2021). The shade also reduces the light available for photosynthesis, and regulators increasingly require AV systems to retain most of the reference yield; the German pre-standard DIN SPEC 91434 sets a floor of 66% (DIN, 2021), and other schemes and farmers set stricter ones. Technology that lets an AV system follow such a floor, by choosing the density of the array and the rotation of its trackers, is therefore a problem of control and design for agriculture.

Single-axis trackers add a degree of freedom: modules can be rotated away from the sun to let light through when the crop needs it (Valle et al., 2017), or towards the sun to maximise electricity. We call the daily choice between the two *light sharing*. It is a sequential decision problem, because the value of light depends on the crop stage, soil water and weather, and the consequence appears at harvest. Learning-based control for crop management has mostly used reinforcement learning with black-box simulators (Gautron et al., 2022), which would need very many simulated seasons for global multi-decadal assessments. Differentiable process models, which fit parameters and policies by gradient descent through a simulator (Reichstein et al., 2019; Shen et al., 2023), are an alternative that has not been applied to AV control. A second difficulty is that the crop response to shade, the weakest part of any AV model, is uncertain, which calls for designs that perform acceptably across plausible models and not only under the best estimate (Marchau et al., 2019).

This paper develops and tests a differentiable-simulation approach to AV tracker control and design, and asks what learning adds. We built a differentiable simulator that couples hourly solar geometry, tracker and PV physics, light and microclimate under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty, and trained a neural controller (SALS) through whole seasons by back-propagation. We compared it with static arrays and with written and tuned rules in a protocol in which the array density of every site is chosen from its own history

only and performance is measured on geographically held-out sites and years. The shade response was learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) and propagated as an ensemble of plausible responses; on that ensemble we tested design margins calibrated by conformal risk control and a learned design network. The protocol was applied to 400 crop–site pairs sampled in proportion to harvested area (Yu et al., 2020), with NASA POWER weather for 2001–2020 (19 sowing years) (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The evaluation is by simulation on public data, without field experiments under trackers. The main findings are as follows.

1. Differentiable simulation makes gradient-based training of a tracker controller through whole seasons tractable (300 steps), but the learned controller adds 0.044 LER over a written phenology rule and only 0.007 over a window optimised on the training seasons, with no difference that can be distinguished from zero at matched retention.
2. Light sharing by rule is the main lever on electricity for single-axis trackers: arrays carry 2.1 to 2.9 times the module area of static arrays at the same mean retention of 90%, at the price of missing the floor in 24–31% of seasons.
3. The shade response is only partly learnable: a hierarchical model fitted to 428 observations predicts a new study to about 0.22 in relative yield, no better than proportional loss for the four staple crop types, and the simulator is milder than the data for maize.
4. A density chosen under the mean response leaves 70% of pairs at the floor when another plausible response holds. A conformal margin attains a stated compliance inside the ensemble, a learned design network does worse than a simple margin, and only large margins are safe outside the ensemble.

2. Materials and methods

The full description, with all equations, parameters and training details, is in the Supplementary Material (Section S1); this section gives what is needed to follow the results.

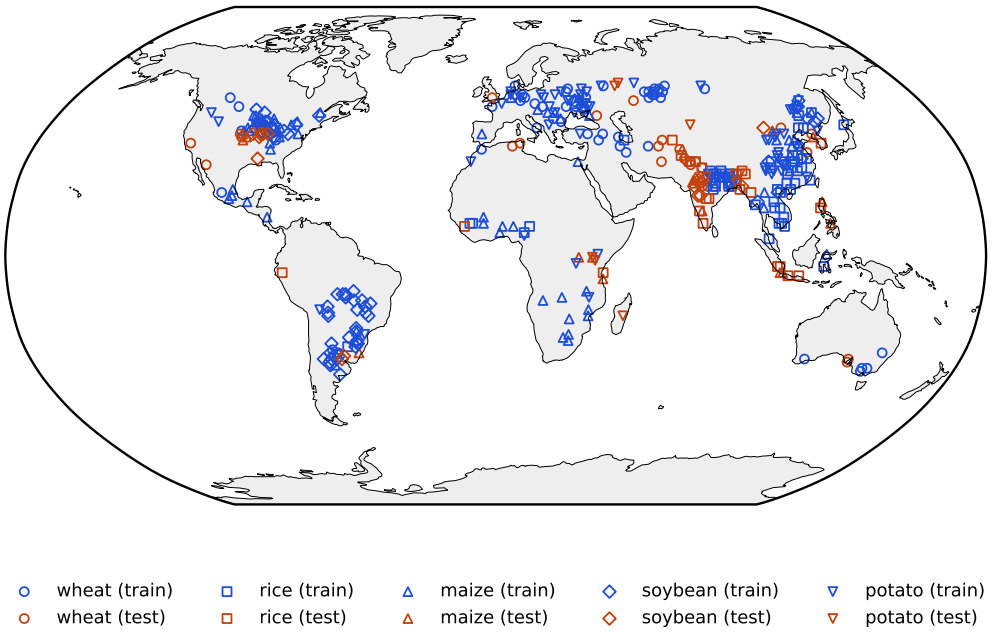


Figure 1: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

2.1. Sites, weather and scenarios

We studied five staple crops (wheat, rice, maize, soybean, potato; about 670 Mha of harvested area). Harvested areas came from SPAM 2010 (Yu et al., 2020). For each crop, 80 one-degree cells with at least 5000 ha were drawn with probability proportional to harvested area, so that an unweighted mean over the 400 crop–site pairs approximates an area-weighted global mean (Fig. 1). Sowing and maturity dates came from the GGCMI Phase 3 crop calendar (Jägermeyr et al., 2021) and daily weather for 2001–2020 from NASA POWER (Sparks, 2018). Each site contributes 19 seasons (7600 site-seasons), each simulated for the 365 days after sowing. Warming levels of +1.5, +2 and +3 °C were imposed by shifting daily temperatures uniformly, with CO_2 of 450, 500 and 600 ppm (390 ppm in the baseline); precipitation and irradiance were unchanged.

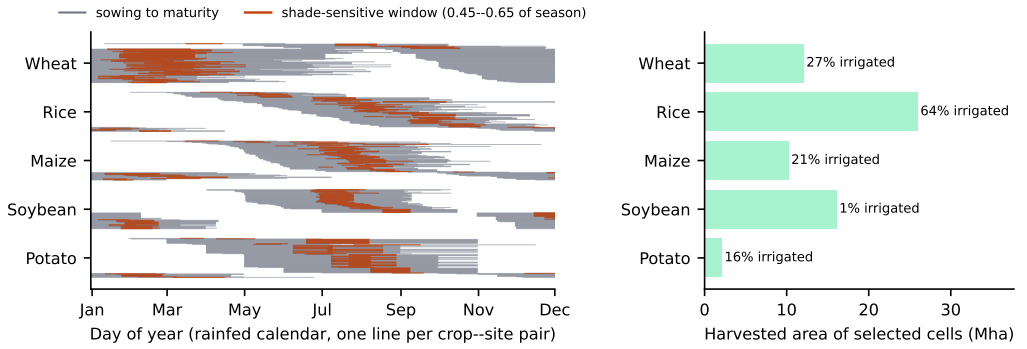


Figure 2: Agronomic data behind the simulations. Left: crop calendar of the 400 crop–site pairs (GGCMI Phase 3 sowing and maturity, one line per pair); the red part marks the window of 0.45 to 0.65 of the season in which the crop model penalises shade through the harvest index. Right: harvested area of the 400 sampled cells (SPAM 2010, not the global area of each crop) and its irrigated share.

2.2. A differentiable agrivoltaic simulator

The simulator (Fig. 3) advances all site-seasons in parallel, with a daily step for the crop and an hourly step for radiation and PV, in PyTorch (Paszke et al., 2019), so that yield and electricity are differentiable with respect to design and control variables. We considered elevated horizontal single-axis trackers with north–south axes, an axis height of 1.25 module widths, a rotation limit of $\pm 60^\circ$ and $g \leq 0.5$. With backtracking rotation ϕ_{bt} (Lorenzo et al., 2011; Marion and Dobos, 2013) and the rotation ϕ_{sh} that minimises the shadow on the crop, a daily *light-sharing level* $u \in [0, 1]$ sets the hourly rotation as

$$\phi = (1 - u)\,\phi_{\mathrm{bt}} + u\,\phi_{\mathrm{sh}}. \tag{1}$$

Plane-of-array irradiance combines beam light reduced by row-to-row shading, isotropic sky diffuse and ground-reflected light; electricity follows from module efficiency (0.21), the NOCT cell temperature and 14% system losses, and a conventional ground-mounted plant ($g = 0.4$) is the energy reference E_{PV}. Beam and diffuse light at the crop follow from the geometry (the diffuse part from a pre-computed two-dimensional sky-view factor), and the canopy maximum temperature is lowered by $\kappa(1 - s)$ with s the share of open-field irradiance that reaches the crop and a moderate $\kappa = 1\,^\circ\mathrm{C}$ (sensitivity: $\kappa = 0$ and $2\,^\circ\mathrm{C}$). Reference evapotranspiration is recomputed under the modules

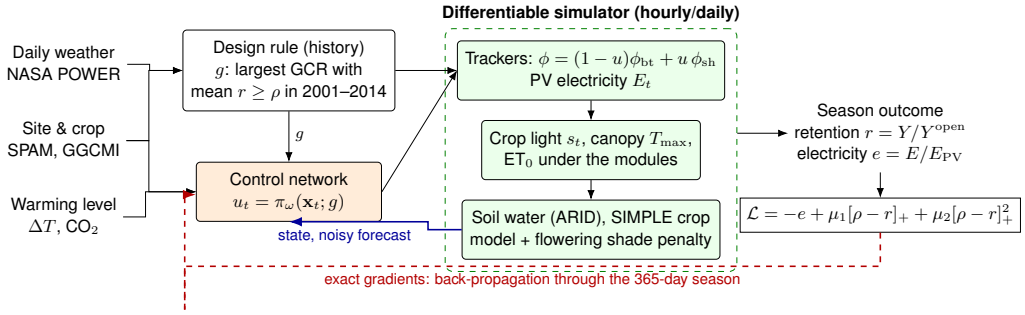


Figure 3: Structure of the simulator. Daily weather and the light-sharing level u determine the tracker rotation, PV electricity, the light and temperature at the crop and the crop state; yield retention and electricity relative to a conventional plant enter the objective.

(Allen et al., 1998). Crop growth follows SIMPLE (Zhao et al., 2019),

$$\Delta B = R_{\mathrm{crop}} \cdot \mathrm{RUE} \cdot f_{\mathrm{Solar}} \cdot f_{\mathrm{CO_2}} \cdot f_{\mathrm{Temp}} \cdot \min\{f_{\mathrm{Heat}}, f_{\mathrm{Water}}\}, \tag{2}$$

with $R_{\mathrm{crop}} = sR$ the radiation at the crop, a soil-water balance for drought stress, a thermal-time requirement calibrated per site from the crop calendar, and each site split into a rainfed and an irrigated sub-field. SIMPLE has a fixed harvest index, so we added a penalty for shade around flowering: in the window $0.45 \leq \mathrm{TT}/T_{\mathrm{sum}} \leq 0.65$ the harvest index is multiplied by $1 - h\bar{d}$, with $\bar{d}$ the mean radiation deficit and a sensitivity h of 0.6 for maize, 0.5 for soybean and rice, 0.4 for wheat and 0.2 for potato (assumptions that reproduce the ranking of crop types in the data). The PV part agrees with PVGIS at low density ($r = 0.97$ at 29 sites, simulator 4.9% lower) and was compared with pvlib (Holmgren et al., 2018) at higher density (Section 3.3).

2.3. Light-sharing control, design and baselines

SALS is a multilayer perceptron (two hidden layers of 64 units) that sets u each day from the crop and soil state, noisy forecasts of maximum temperature, irradiance and ET_0 (errors of 2 °C, 20% and 15%), the irrigation status, the density, the latitude, the day of season and the crop. It is trained by back-propagation through the 365 daily steps of the simulator to minimise $-e + \mu_1 \max\{0, \rho - r\} + \mu_2 \max\{0, \rho - r\}^2$ with r the yield retention, e the electricity relative to a conventional plant and $\rho = 0.9$, after behaviour cloning of the phenology rule (trained from random weights it converges to the seasonal-sharing rule). Its gain over the phenology rule therefore measures what fine-tuning through the simulator adds to that rule.

Every method uses the same *history-based design rule*: the density of a site is the largest value on the grid (step 0.025 in the main split, 0.05 in the pooled analyses) for which the mean retention over its 2001–2014 seasons is at least 0.9; if none qualifies, the site stays open field. Test seasons (2015–2019) are never used for design, and test sites never for training. We compare static arrays (backtracking, $u = 0$), seasonal sharing ($u = 1$ throughout the season), a stress rule (sharing except on forecast heat or drought days), a written phenology rule (sharing for $0.05 \leq \mathrm{TT}/T_{\mathrm{sum}} \leq 1$), a window tuned on the training seasons of each split (grid-tuned and CMA-ES optimised (Hansen and Ostermeier, 2001); Supplementary Material) and SALS, with a conventional plant and the open field as references.

2.4. Shade response: data, ensembles and design under uncertainty

The crop response to shade was learned from 428 observations of 58 studies (Laub et al., 2022) as $y(s) = \exp(-\beta s^{\gamma})$ with partial pooling over crop types and compared by leave-one-study-out cross-validation with proportional loss $y = 1 - s$ and with gradient-boosted trees. The simulator has two shade parameters per crop, a radiation-use compensation and a multiplier on the harvest-index penalty. Because the simulator is differentiable, they were fitted by gradient descent through whole seasons to the learned curves (20% and 40% shading) and to field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023). Repeating the fit on randomly perturbed observations gave an ensemble of plausible responses (E2; the Supplement also uses a wider ensemble E3). The members differ by a standard deviation of 0.02–0.06 in relative yield, generally less than the between-study error of 0.22.

The density under uncertainty is chosen as (i) the *mean-response design*, the design rule under the mean of the members; (ii) a *margin*, the same rule with the floor raised to $d > 0.9$ under the mean response; (iii) an *ensemble rule* that requires the floor in a share of the members; (iv) a *conformal margin*, the smallest d whose conformal-adjusted risk of missing the floor, $n/(n+1)\,\hat{R}(d) + 1/(n+1)$, is at most a target α (Angelopoulos et al., 2024), with the pair as the unit, calibrated on the pairs of two splits that are not in the third and tested on the third; and (v) a *learned design network*, a small multilayer perceptron that chooses a density from the history retention curves of the other members (mean, minimum and maximum at each density), the crop and the latitude, trained to minimise $-\mathrm{LER} + \lambda\,\mathbf{1}[r < 0.9]$ under the held-out member, with the same leave-one-split-out evaluation (details in the

Supplement). Pairs held out in several splits are averaged. The truth is a held-out member, or one of four fixed responses (proportional, calibrated, field-fitted and harsh; only the field-fitted lies within the range of the ensemble) with the belief the mean of the ensemble's 12 members, and an oracle design under the true response is the reference.

2.5. Evaluation

Cost and scale are summarised with an illustrative annual net value per hectare relative to the open field (assumed parameters, reported as break-even prices and sensitivities; Supplement). Sites were split spatially (10° blocks, 70% training, 30% test) with three splits; training used 2001–2014 and evaluation the 2015–2019 seasons of held-out sites, and results are for 92 held-out pairs (main split) or 76 distinct pairs pooled over three splits. Metrics are electricity, mean and production-weighted retention, compliance (share of seasons or sites with retention ≥ 0.9), and the land-equivalent ratio $\mathrm{LER} = r + e$ (Dupraz et al., 2011), relative to the open field of the same climate. Intervals are cluster bootstrap intervals over crop–site pairs; SALS was trained with three seeds per split. For the global estimate, each crop's mean electricity per hectare is scaled to its physical area in SPAM 2010.

3. Results

Further analyses are in the Supplementary Material (Section S2): other warming levels, floors, canopy cooling, forecast errors, layouts, controllers trained on ensembles of responses, and the cross-validation tables of the shade-response model.

3.1. Light sharing under a yield floor

Table 1 compares the controllers on the 92 held-out crop–site pairs of the main split, after the density of each site has been chosen from its 2001–2014 seasons with the same rule for all controllers; all methods therefore keep the mean retention close to 90%, and differences in electricity are differences in the density that each controller can sustain. Without light sharing the floor limits ground coverage to 6% and arrays produce $217\,\mathrm{MWh\,ha^{-1}\,yr^{-1}}$ (LER 1.07). Sharing light during the cropping season allows 17% coverage and $397\,\mathrm{MWh\,ha^{-1}\,yr^{-1}}$ (LER 1.19), and a written rule that follows crop phenology reaches 465 (LER 1.24). A stress rule that shades the crop on forecast

Table 1: Held-out performance (92 test pairs, seasons 2015–2019) under the baseline and the +3 °C climate. The density (GCR) of each site is the largest value on a grid with step 0.025 at which the method's mean retention in the site's 2001–2014 seasons is ≥ 0.9. E: electricity (MWh ha^{-1} yr^{-1}); e: electricity relative to a conventional plant; r: mean yield retention; r_{prod}: production-weighted retention; C_s, $\mathrm{C}_{\mathrm{site}}$: share of seasons and of sites with retention ≥ 0.9 (%); LER $= r + e$. Other warming levels are in Supplementary Table S2.

Method	GCR	E	e	r	r_{prod}	C_s	$\mathrm{C}_{\mathrm{site}}$	LER
Baseline								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS	0.17	510	0.36	0.905	0.902	70	74	1.27
+3 °C								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS	0.18	540	0.39	0.907	0.904	68	75	1.30

heat or drought days is worse (LER 1.18), because it cannot sustain the density. SALS sustains 17% coverage, produces 510 MWh ha^{-1} yr^{-1} and reaches an LER of 1.270; this is +0.038 over the phenology rule (site-mean contrast, 95% interval 0.025–0.053, higher at 84% of the 92 sites; the season-pooled values of Table 1 give 1.270 against 1.236) and +0.205 (0.176–0.236) over static arrays, with 2.35 times the electricity at 2.9 times the density (Fig. 4). The ranking, SALS ahead of the phenology rule, ahead of the stress rule and seasonal sharing, ahead of static arrays, holds in every warming scenario (SALS LER 1.30 at +3 °C), for floors of 0.80 to 0.95 (SALS minus phenology rule 0.033–0.037 LER), for canopy cooling of 0, 1 and 2 °C (0.029, 0.034, 0.042), under larger forecast errors and under alternative layouts (Supplement). Masking the soil-water, aridity and forecast inputs of SALS leaves its LER unchanged (1.269 against 1.270). The price of light sharing is compliance: the floor is met in 69–76% of seasons, against 93% for static arrays, which also keep a slightly higher mean retention (0.917 against about 0.905) because the density grid forces a coarser choice at low density.

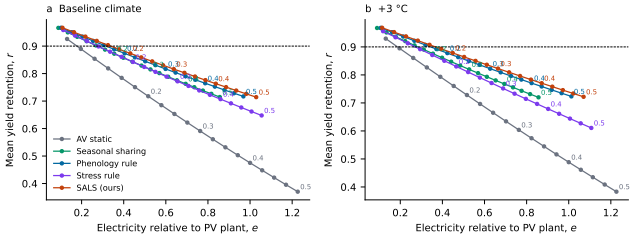


Figure 4: Land-equivalent ratio against ground coverage ratio for the controllers on the 92 held-out pairs: at matched densities, SALS lies on or slightly above the phenology rule, both above seasonal sharing and the stress rule, and static arrays are far below.

Benefits differ between crops: soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; LER 1.41 and 1.30), potato follows (0.19; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; LER 1.20 for both). The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize because high-yielding seasons lose more, and regionally it is lowest in Latin America and Oceania (0.862, 95% site-bootstrap interval 0.841–0.897, four sites, GCR 0.41) and Africa and the Middle East (0.894, 0.868–0.908, eight sites), against 0.902 for South Asia (39 sites). Only the first deficit lies outside the interval of South Asia, so the regional comparison is an indication and not a finding.

3.2. What learning adds

The written phenology rule was not tuned, whereas SALS is trained, so we tuned its window on the training seasons of each split, by a grid search and by CMA-ES over a five-parameter schedule. Pooled over three splits and 76 distinct pairs (SALS: mean of three training seeds per split), tuning the window raises LER over the written rule by 0.026 (grid; 95% cluster-bootstrap interval 0.016–0.037) and 0.036 (CMA-ES; 0.023–0.052), and SALS by 0.044 (0.030–0.060) (Table 2). Optimising a window therefore accounts for about 83% of the margin of SALS over the written rule. The optimum is an almost binary window from 0.20 to 0.93 of the season, so a smooth ramp is not needed to reproduce most of what SALS does. SALS exceeds the CMA-ES window by 0.007 LER (0.002–0.014), or 2.4% more electricity at the designed densities; the margin is positive on each split (0.008, 0.009 and 0.005, with the interval of the third including zero) and for each of the three training seeds (0.007, 0.008, 0.007). The design step quantises density to a grid of 0.05, much coarser than this margin, so we also compared

Table 2: Rules and the learned controller (SALS: mean of three training seeds per split) pooled over the three spatial splits (76 distinct crop–site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The grid-tuned window shares light while thermal time lies between 0.15 and 1.0 of the season requirement (best of a 4×3 grid on the training-block seasons of each split, the same on all three; the written rule uses 0.05 to 1.0). The CMA-ES schedule has five parameters (start, end, two ramp widths, level) and was tuned on each split's training block and stopped, at the generation cap, at an almost binary window from 0.20 to 0.93 of the requirement at level 1.00 (see Methods).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule (written)	0.154	0.292	0.916	85	91	1.207
Grid-tuned window	0.161	0.317	0.916	84	89	1.233
CMA-ES optimised window	0.158	0.327	0.916	83	87	1.244
SALS	0.161	0.335	0.916	83	87	1.251

the controllers at matched retention, reading LER at a retention of 0.90 off each pair's retention–LER curve: SALS exceeds the CMA-ES window by 0.003 (95% interval -0.0004 to 0.007; 73 pairs whose retention curves bracket 0.90), which is not distinguishable from zero. The source of the remaining margin is not identified: it does not come from soil water, forecasts or aridity, and canopy development and density remain untested. Training SALS on ensembles of shade responses or giving it a farm-measurable biomass signal did not help within our budget (Supplement), and SALS-robust was below SALS even under responses inside its training ensemble, which may point to under-training.

3.3. Validation and the shade response

PV model. The simulated specific yield of an isolated row agrees with PVGIS at 29 sites ($r = 0.971$, simulator 4.9% lower). At higher ground coverage we compared plane-of-array irradiance with pvlib (Holmgren et al., 2018) (infinite sheds, isotropic sky, backtracking, same hourly irradiance, 12 sites): the simulator is 0.4% above pvlib at $g = 0.05$ and 0.7, 1.4, 2.1, 2.6 and 3.0% above at $g = 0.1$ to 0.5, so it underestimates the loss relative to a sparse array (2.8% instead of 4.4% at $g = 0.3$). Correcting the electricity of the arrays and of the reference plant by this ratio raises the LER of light-sharing methods by 0.003–0.004 (0.008 with the squared ratio), leaves the margins of SALS unchanged within 0.0003, raises break-even prices by about

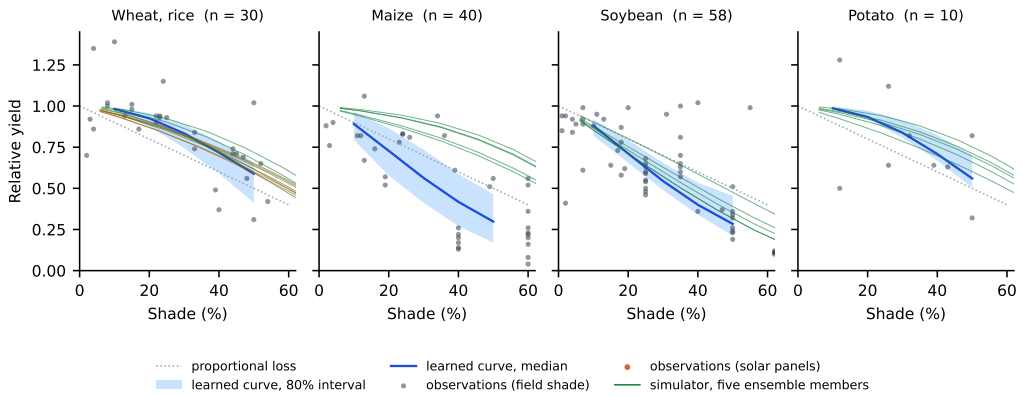


Figure 5: Yield response to shade: observations of the meta-analysis (Laub et al., 2022) (grey: field shade; orange: solar-panel shade), the hierarchical curve learned from them (median and 80% bootstrap interval over studies) and the simulated response of each member of ensemble E2 (green; wheat and rice in green and brown). The curve of C3 cereals stands for rice, which the data set lacks. For maize the simulator is milder than the data.

0.8 USD MWh^{-1} and changes no ranking. The comparison is model-to-model and does not test the ground-side light.

Shade response. Fig. 5 shows the observations of the meta-analysis (Laub et al., 2022), the learned curves and the simulator. In leave-one-study-out cross-validation over all 339 shaded observations a hierarchical crop-type model reduces the error from 0.43 (proportional loss) to 0.31, but this gain comes from crop types that the simulator does not represent. For the four staple crop types the learned curves are no better than proportional loss (0.222 against 0.220; difference +0.002, interval −0.023 to +0.030), and gradient-boosted trees are worse. The error of 0.22 is the between-study spread that a design has to cover. The simulator is close to the data for cereals and potato, and conservative against the field trials of rice, wheat and potato (for example rice 0.79 against 0.84 (Gonocruz et al., 2021)), but milder than the meta-analysis for maize (0.75 against 0.42 at 40% shading); for soybean its value (0.48) lies within the interval of the data (0.40, 80% interval 0.34–0.53). One maize trial reports smaller losses still (Sekiyama and Nagashima, 2019), so for maize the direction of the bias is uncertain.

3.4. Design under shade-response uncertainty

Table 3 and Fig. 6 use ensemble E2 (five members that differ by 0.02 to 0.06 in relative yield, against the between-study error of 0.22). The truth is a held-out member and the design belief is the mean of the other members; all numbers are for 76 distinct pairs, with a pair held out in several splits averaged. A density chosen under the mean response leaves 70% of pairs at the floor, against 87% for an oracle design under the true response; the oracle's LER (1.486) is slightly below that of the mean-response design (1.493), which obtains it by missing the floor more often. Raising the floor used in the design to 0.92 gives 88% at 0.067 less LER, 0.94 gives 97% and 0.96 gives 99%, and an ensemble-quantile rule lies on the same frontier as the margin (Supplement), so the ensemble is useful to size a margin and not to replace it. Compliance by crop is uneven (54–80% under the mean response, 80–98% for a margin of 0.92).

Conformal margins. Conformal risk control (Angelopoulos et al., 2024) picks the smallest margin whose adjusted estimate of the average share of members at which a pair misses the floor is at most a target, which bounds the expected share of misses for a new pair if pairs are exchangeable. We calibrated it on the pairs of two splits that are not in the third split (margins of 0.915–0.930) and evaluated it on the third. The pooled compliance of the test pairs was 94, 89 and 86% for targets of 90, 85 and 80%, and 82.5–97.1% across splits; the largest shortfall below a target was 2.5 points (split 0, target 85%), within the sampling error of about 5 points for 26–35 pairs per split. The guarantee concerns the average over members, not each member, and it is not available per crop, because there are only 3 to 9 calibration pairs per crop and split: a single margin gives 78–100% by crop, and we did not find evidence that per-crop margins help.

Learned design. A neural network (one hidden layer of 32 units) that chooses the density from the retention curves of the other members (mean, minimum and maximum at each density), the crop and the latitude, trained for each penalty λ to minimise $-\mathrm{LER} + \lambda\,\mathbf{1}[r < 0.9]$ on the pairs of two splits (three seeds; test-split pairs excluded), does not improve on the margin: at matched compliance its LER is lower than that of the margin frontier by 0.02–0.06 for compliance between 75 and 91% (for example -0.057, 95% interval -0.076 to -0.032, at 87%), and the difference is not distinguishable from zero above 96%. The comparison favours the margins, whose single parameter is read off the test results, but the network has no advantage that this could hide.

Table 3: Design rules for the density under shade-response uncertainty (ensemble E2, phenology rule, 76 distinct crop–site pairs): share of pairs (%) at the floor (mean retention ≥ 0.9, averaged over the held-out members or responses) and mean LER, when the true response is a held-out member (inside) or one of four fixed responses (outside the belief of the ensemble); the third column gives the range over crops of the inside share. Conformal margins and the network are fitted on the pairs of two splits that are not in the third and evaluated on the third (leave-one-split-out); fixed margins and the oracle are not fitted.

	Inside the ensemble			Outside	
Rule	Pairs at floor	LER	By crop	Pairs at floor	LER
Oracle (design under the true response)	87	1.486	–	88	1.283
Design under the mean response	70	1.493	54–80	27	1.435
Fixed margin 0.92	88	1.426	80–98	34	1.376
Fixed margin 0.94	97	1.345	90–100	56	1.303
Fixed margin 0.96	99	1.251	96–100	73	1.220
Conformal margin, target 90%	94	1.396	85–100	37	1.350
Conformal margin, target 85%	89	1.424	81–97	34	1.373
Conformal margin, target 80%	86	1.440	78–96	32	1.390
Learned design network, $\lambda = 0.8$	84	1.401	–	35	1.345
Learned design network, $\lambda = 1.0$	87	1.378	–	36	1.328
Learned design network, $\lambda = 1.5$	94	1.342	–	41	1.295
Learned design network, $\lambda = 2.0$	97	1.330	–	43	1.285

Outside the ensemble. The truth was also one of four fixed responses (proportional, calibrated to the meta-analysis, field-fitted and harsh), with the design belief the mean of the 12 members of E2; only the field-fitted response lies within the range spanned by the ensemble, and the pooled values average the four, so a value of 25% for a crop means that one of four responses is met. No rule is safe: conformal margins reach 32–37%, the network 35–43%, a margin of 0.96 73% at a cost of about 0.2 in LER relative to the mean-response design, and the oracle 88%. Wheat, rice and potato pairs are at the floor in 3–29% of cases for the mean-response design, the margin of 0.92 and the conformal margins, whereas soybean pairs meet it in 90–100%, because the ensemble already contains steep soybean responses (Supplement). The guarantees hold only within the range of responses that the ensemble spans.

3.5. Cost, warming and scale

Cost, warming and scale are secondary to the technology questions and are treated as sensitivity analyses in the Supplementary Material (Section S2).

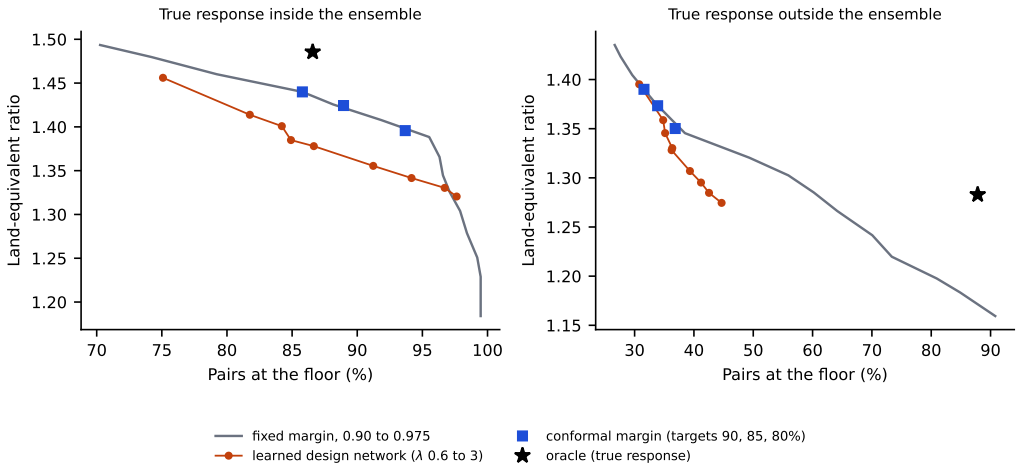


Figure 6: Land-equivalent ratio against the share of pairs at the floor for fixed margins, conformal margins (global and per crop), the learned design network (at several penalties λ) and the oracle, when the true response is a held-out member of the ensemble (left) or a fixed response outside it (right).

With illustrative costs (Supplementary Table S9), the light-sharing methods break even with the open field at 70 to 92 USD MWh^{-1} (SALS 73, 95% interval 69–76) against 59 for static arrays, and the difference between SALS and static arrays at 90 USD MWh^{-1} is not distinguishable from zero; a conventional plant that replaces the crop breaks even at 48 and is more profitable than every AV option above that price, a result that rests on an assumed capex 25% below the AV structure (with equal capex it breaks even at about 63). Under +3 °C the open-field production of the held-out sites falls by 23% for maize and 18% for rice, AV production follows the open field at about 90% of its level, and the LER, which is relative to the open field of the same climate, rises slightly (SALS 1.270 to 1.297). The retention is lowest in Latin America and Oceania (0.862, 95% interval 0.841–0.897, four sites), whereas the interval of Africa and the Middle East (0.868–0.908) overlaps that of South Asia (0.902), so the regional comparison is an indication. Scaled to the physical area of the five crops, AV on 5% of the world's staple cropland would supply about 14 PWh yr^{-1}, a field-scale technical potential without an interval that needs about 10.6 TWp of modules, about half of it from maize and soybean, whose shade response is the least certain.

4. Discussion

4.1. What the results establish

Sharing light with the crop is the main lever on the electricity that single-axis tracker arrays can produce under a strict yield floor (east–west, vertical and fixed-tilt layouts were not tested): with static trackers the floor limits arrays to about 6% ground coverage (LER 1.07), and any light-sharing control allows 2.1 to 2.9 times the density (LER 1.18–1.24 for written rules). The best simple rule follows crop phenology, needs only the crop's thermal time, which is observable on the farm, and is insensitive to forecast errors, whereas a rule that reacts to heat and drought stress is worse in every climate. The early hypothesis that stress-aware control would be a main source of benefit under warming was not supported: a learned controller without stress inputs performs as well as one with them, and agrivoltaics does not protect staple production against warming in the model, because the systems produce about 90% of the open-field yield in every climate and follow the same decline in maize and rice. These results are for the calibrated shade response of the main analysis; the ranking of the methods also holds under four alternative responses (tested on 21 pairs with one seed; Supplement), but the gains for maize and soybean, which supply about half of the upscaled electricity, are the least certain.

4.2. What learning adds

We tested four ways in which learning could matter, and each gives a small or null return. *Learning the shade response from data* is useful mainly as a measurement of how little can be learned: a hierarchical model fitted to 428 observations predicts a held-out study of the four staple crop types to 0.22 in relative yield, no better than proportional loss, and its gain over proportional loss comes from crop types that the simulator does not represent. The response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve. *Calibration through the differentiable simulator* produced an ensemble of plausible responses that fits the trials and the curves for cereals and potato but not those of maize, and it quantifies the design risk. *Control* gains little: a network of 5441 weights trained through whole seasons exceeds the written rule by 0.044 LER, but a window optimised by a derivative-free search (five parameters, 200 simulator runs) is only 0.007 LER behind it, and at matched retention the difference is not distinguishable from zero; we did not test whether such a search would

work on the network itself. A reader who needs a simple, auditable controller can use the window and obtain about 83% of the gain. *Learning the design margin* gave a guarantee but no better frontier: conformal calibration attains a stated compliance within the ensemble, with margins that can be set per crop, and a neural design policy does no better than a simple margin, and neither is safe outside the ensemble. The honest conclusion for the use of artificial intelligence here is that differentiable simulation makes calibration, control and design tractable, and that the data, not the learner, limit the result: the shade response that enters the design is uncertain by more than the differences between controllers, so that the choice of design belief matters more for compliance than the choice of controller.

4.3. Implications for control and design in agriculture

Holding the mean retention at 90% does not hold it in each season or at each site. About 30% of seasons with light-sharing control fall short of the floor on the main split, the shortfall is mostly small (2.7% of SALS seasons are below 0.8 of the open-field yield), and production-weighted retention is below 0.9 for soybean and maize and in several regions. A floor enforced per site would therefore need a crop- and site-specific design step, and a margin that is calibrated rather than chosen by hand. Compliance is sensitive to the response assumed in the design: a density chosen under the mean response leaves 70% of pairs at the floor when another plausible response holds, a margin or a conformal calibration repairs this inside the ensemble, but only a margin of 0.96, at a cost of about 0.2 LER, protects most pairs against responses outside it, notably for cereals and tubers when the ensemble is anchored on mild responses. Because all controllers are designed to the same mean retention, their compliance is close by construction, so the choice of controller and the choice of the design belief are different decisions. The farmer's exposure is mainly one of compliance with the floor, and it concentrates where the energy gain is largest, in rainfed maize and soybean areas, where the shade response is least certain; the economic comparison with a conventional plant (Supplement) depends on assumed costs.

4.4. Limitations

All results come from a simulator. The PV model agrees with PVGIS at low density and with pvlib to within 3% of plane-of-array irradiance up to a ground coverage of 0.5, but it underestimates the shading loss of dense arrays by 1.5 to 2.5 points, which changes the LER by at most 0.008 and

no ranking, and no measured array output at these densities was used. The shade response was learned from one meta-analysis and compared with four field trials; for maize the data and the simulator disagree (0.42 against 0.75 at 40% shading), the harvest-index sensitivities are assumptions, and the sensitivity analyses use subsamples of pairs. The floor of 0.9 is a modelling convention and not a regulatory threshold (DIN SPEC 91434 uses 66%). Canopy cooling is one coefficient applied to a heat-stress function calibrated on air temperature; SIMPLE has a fixed harvest index and no emergence, early canopy closure, weeds or machinery effects; the climate scenarios are deltas that isolate the thermal signal; and the layouts are single-axis trackers over flat fields, with no east–west, vertical or fixed-tilt arrays. The economic analysis uses assumed parameters and a plant cheaper per watt than the agrivoltaic structure, and omits land rent, degradation, extra structure cost, curtailment and subsidies. The conformal and learned design results come from five ensemble members and 76 pairs, with split-level deviations of up to three points from the stated compliance, and the controller comparisons use one network per training seed. The regional and global figures are indications and a scaling convention without intervals. Future work should validate the shade response under trackers for maize and legumes, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers and designs that guarantee the floor per site and per season.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever (by rule, not by learning) on the electricity of single-axis tracker agrivoltaics at a strict yield-retention floor of 90% (other layouts were not tested). Evaluated on held-out crop–site pairs, with the density of each site chosen from its own history, light-sharing rules allowed 2.1 to 2.9 times the module density of static arrays at a mean retention of 90% and raised the land-equivalent ratio from 1.07 to 1.18–1.24. A neural controller trained through a differentiable simulator exceeded the written phenology rule by 0.044 LER but a window optimised by CMA-ES by only 0.007, and by an amount not distinguishable from zero at matched retention; ensemble training and a biomass signal did not help.

The crop response to shade is the least certain input. A model fitted to 428 observations predicts a new study to about 0.22 in relative yield, no

better than proportional loss for the staple crop types, and the simulator is milder than the data for maize. A density chosen under the mean response left about 70% of pairs at the floor when another plausible response held, against 87% for an oracle design. Conformal calibration attained the stated compliance inside the ensemble (within 2.5 points per split), a learned design network did worse than a simple margin, and no rule short of a margin of 0.96 was safe outside the ensemble.

With illustrative costs, light sharing breaks even with the open field at 70 to 92 $\mathrm{USD\,MWh^{-1}}$ at floor-designed densities, and a conventional plant replacing the crop is more profitable above 48 under an assumed lower capex (Supplement); the field-scale electricity potential of 5% of staple cropland (about 14 $\mathrm{PWh\,yr^{-1}}$) is a technical potential without an interval. For agricultural decision-making, the practical message is to treat the crop response as a distribution, to calibrate the design margin per crop instead of choosing it by hand, and to expect that only large margins protect against responses outside the range considered.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al., 2020), the GGCMI Phase 3 crop calendar (Jägermeyr et al., 2021), NASA POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author's public repository (`https://github.`

com/vinhqdang/Artificial-Intelligence-in-Agriculture); a DOI-registered archive of the code and results will be deposited and cited here before publication.

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